Inspiration Private School
MHF4U

Chapter #6 & 7 Summative Test

Name:____________________________ Date:____________________________

Knowledge	Application	Thinking	Communication
/20	/12	/5	/4

KNOWLEDGE

Multiple choice (4 marks)

1. $\sqrt{8}$ is equivalent to

a. 4 b. $2^{\frac{2}{3}}$ c. $2^{\frac{3}{2}}$ d. $\frac{1}{2^3}$

2. What is the solution to $2^{x+1} = 4^x$

a. 1 b. 2 c. 4 d. -1

3. What is the product law of logarithm?

a. $log\, a + \log b = \log(a + b)$
b. $\log a \bullet \log b = log\,(a + b)$
c. $\log a \bullet \log b = \log(ab)$
d. $\log a + \log b = \log(ab)$

4. The expression $log\,5 + log\,10 - log\,25$ is equal to

a. 1 b. log5 c. 0 d. log2

5. Convert from exponential to logarithmic form (1 mark)

$$3^5 = 243$$

6. Convert from logarithmic to exponential form(1 mark)

$$1296 = 4^x$$

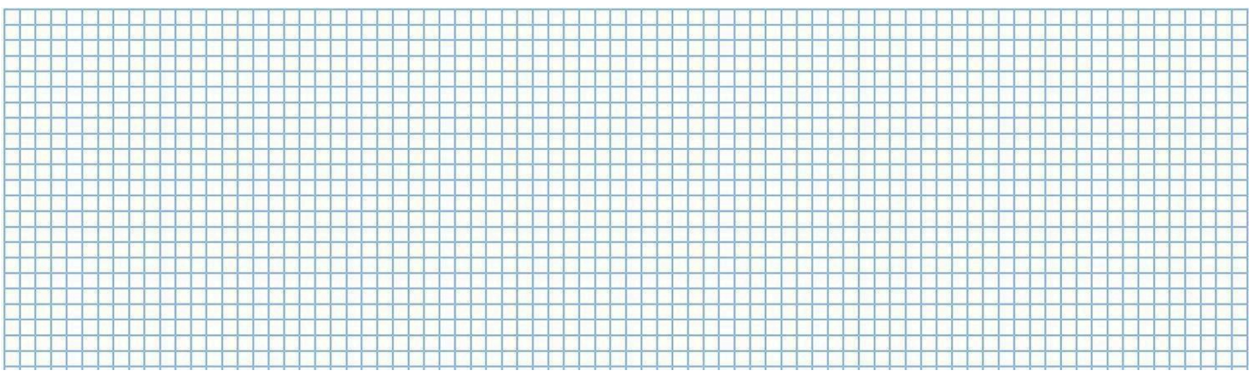
7. Express $log(x^2) + log(3y^{\frac{1}{3}}) - log(x)$ as a single logarithm. (2 marks)

8. For the function $y =- 2log(2x - 4) + 2$, determine the following (6 marks)

a. Horizontal compression:_____________ b. Horizontal translation:_________________

c. Asymptote:________________________ d. x-intercept:__________________________

e. Sketch the function:



9. Solve each of the following equations. Show your work (6 marks)

a. $4^{2x} - 4^{x} - 20 = 0$

b. $64^{x+5} = 16^{2x-1}$

c. $2^{x} + 12(2)^{-x} = 7$

APPLICATION

10. Solve each of the following equations. Show your work. (6 marks)

a. $\log\ (x + 5)\ - log\ (x + 1)\ = \log\ (3x)$

b. $\log(x - 6\ ) = 1 - log(x - 2)$

11. When you consume caffeine, the percent P, of caffeine in your bloodstream is related elapsed time t, in hours by $t = 5\left(\frac{\log log\ P}{\log log\ 0.5}\right)$ (6 marks)

a. How long will it take for the amount of caffeine the drop to 20% of the amount consumed?

b. Suppose you drink a cup of coffee at 9:00 am, what percent of the caffeine will remain in your body at noon?

THINKING

12. If $v = 27$ and $w = 5$, write 225 in terms of v and w (5 marks)

COMMUNCATION

13. How are exponential and logarithmic functions related? Discuss verbally with the teacher. (2 marks)

14. How would you solve for *t* in the equation $10 = 2^t$? (2 marks)